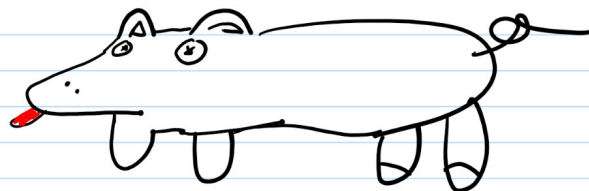


Bài tập GTL



1) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 2x + 5} - x)$

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 2x + 5 - x^2}{\sqrt{x^2 + 2x + 5} + x} = \lim_{x \rightarrow +\infty} \frac{x(2 + \frac{5}{x})}{|x|\sqrt{1 + \frac{2}{x} + \frac{5}{x^2}} + x} = \lim_{x \rightarrow +\infty} \frac{2x}{2x} = 1$$

2) $\lim_{x \rightarrow -\infty} (\sqrt{x^2 - 5x - 1} - \sqrt{x^2 + 3x + 3})$

$$\lim_{x \rightarrow -\infty} \frac{-8x - 4}{\sqrt{x^2 - 5x - 1} + \sqrt{x^2 + 3x + 3}} = 4$$

3) $\lim_{x \rightarrow 0} \frac{\sqrt{\cos x} - \sqrt[3]{\cos x}}{\sin^2 x}$

$$\lim_{x \rightarrow 0} \left(\frac{\sqrt{\cos x} - \cos x}{1 - \cos^2 x} + \frac{\cos x - \sqrt[3]{\cos x}}{1 - \cos^2 x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\cos x (1 - \cos x)}{(1 - \cos x)(1 + \cos x)(\sqrt{\cos x} + \cos x)} + \lim_{x \rightarrow 0} \frac{-\cos(1 - \cos^2 x)}{(1 - \cos^2 x)(\cos^2 x + \cos x \sqrt{\cos x} + \sqrt{\cos^3 x})}$$

$$= \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}$$

4) $\lim_{x \rightarrow 1} \left(\frac{3}{1 - \sqrt{x}} - \frac{2}{(1 - \sqrt[3]{x})} \right)$

Đặt $t = \sqrt{x} \Rightarrow x = t^2 \rightarrow t \rightarrow 1$

khi đó:

$$\lim_{t \rightarrow 1} \left(\frac{3}{1 - t^2} - \frac{2}{1 - t^3} \right) = \lim_{t \rightarrow 1} \frac{3 - 3t^2 - 2 + 2t^3}{(1 - t^3)(1 - t^2)} = \lim_{t \rightarrow 1} \frac{2t^3 - 3t^2 + 1}{(1 - t)^2(1 + t)(t^2 + t + 1)}$$

$$= \lim_{t \rightarrow 1} \frac{(1 - t)^2(2t + 1)}{(1 - t)^2(1 + t)(t^2 + t + 1)} = 1$$

"Còn đây là chuyện -vũ"
+ exp: vô định là vô số giờ xét.

$$\frac{(x-1)^2(2x+1)}{(1-x)^2(1+x)(x^2+x+1)}$$

$$\begin{cases} x=1 \\ x=-\frac{1}{2} \end{cases}$$

$$5) \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x-1} + \frac{1}{x+1} \right) = \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{x-1+x+1}{x^2-1} \right) \\ = \lim_{x \rightarrow 0} \left(\frac{1}{x} \cdot \frac{2x}{(x^2-1)} \right) = -2$$

$$6) \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + \sqrt{x}}{\sqrt{x} + 1} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \cdot \sqrt{1 + \frac{1}{x}}}{\sqrt{x} \sqrt{1 + \frac{1}{x}}} = 1$$

$$7) \lim_{x \rightarrow \infty} x^2 \left(1 - \cos \frac{1}{x} \right) \quad t = \frac{1}{x} \rightarrow t \rightarrow 0$$

$$\lim_{t \rightarrow 0} \frac{(1 - \cos t)}{t^2} = \lim_{t \rightarrow 0} \frac{t^2}{2t^2} = \frac{1}{2}$$

$$8) \lim_{x \rightarrow 0} \frac{\sqrt{1+2x^2} - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\sqrt{1+2x^2} + \frac{x^2}{2} - 1}{x^2}$$

$$* \sqrt{1+2x^2} - 1 \sim \frac{2x^2}{2}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 + \frac{x^2}{2}}{x^2} = \frac{3}{2}$$

$$9) \lim_{x \rightarrow 0} \frac{\sqrt{5} - \sqrt{4 + \cos x}}{x^2} = \lim_{x \rightarrow 0} \frac{1 - (0)x}{x^2 (\sqrt{5} - \sqrt{4 + \cos x})} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2 (\sqrt{5} + \sqrt{4 + \cos x})} \\ = \frac{1}{4\sqrt{5}}$$

$$10) \lim_{x \rightarrow 2} \frac{2^x - x^2}{x - 2} = \lim_{x \rightarrow 2} \frac{2^x \cdot \ln 2 - 2x}{1} = 4 \ln 2 - 4$$

$$11) \lim_{x \rightarrow 0} \frac{e^{x^3} - 1 + x^2}{x \tan x} = \lim_{x \rightarrow 0} \frac{e^{x^3} - 1 + x^2}{x^2}$$

$$\text{Vi} \quad e^{x^3} - 1 \sim x^3, x \rightarrow 0$$

$\Rightarrow e^{x^3} - 1$ là VCB bậc cao hơn x^2

$$\lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1$$

$$12) \lim_{x \rightarrow 0} \frac{e^x - \cos x}{\sqrt{1+2x} - 1} = \lim_{x \rightarrow 0} \frac{e^x - 1 + 1 - \cos x}{\sqrt{1+2x} - 1}$$

$$\text{Vi} \quad \begin{cases} e^x - 1 \sim x, x \rightarrow 0 \\ 1 - \cos x \sim \frac{x^2}{2}, x \rightarrow 0 \end{cases}$$

$$\Rightarrow 1 - \cos x = o(e^x - 1)$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+2x} - 1} = \lim_{x \rightarrow 0} \frac{(\sqrt{1+2x} + 1)x}{1+2x - 1} = \lim_{x \rightarrow 0} \frac{\sqrt{1+2x}}{2} = \frac{1}{2}$$

$$13) \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sqrt[4]{1+4x^2} - 1} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{2}}{\frac{1}{4} 4x^2} = \frac{1}{2} \quad \begin{aligned} \cos 2x &= 1 - 2\cos^2 x \\ &= 2\sin^2 x - 1 \end{aligned}$$

$$14) I = \lim_{x \rightarrow 0} \frac{1 - \cos 5x}{\ln(1+x \sin x)} = \lim_{x \rightarrow 0} \frac{25x^2}{2 \ln(1+x^2)} = \lim_{x \rightarrow 0} \frac{25x^2}{2x^2} = \frac{25}{2} \quad \begin{aligned} & \cos x \cdot (1 - 2\cos^2 x) \\ & 1 - \cos x \cdot (\cos 2x) \end{aligned}$$

$$15) I = \lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos x} \cdot \sqrt{\cos 2x}}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{1 - 2\cos^3 x + \cos x}{(1 - \cos^2 x)(1 + \sqrt{\cos x} \cdot \sqrt{\cos 2x})}$$

$$\lim_{x \rightarrow 0} \frac{(\cos x - 1)(-2\cos^2 x - 2\cos x - 1)}{(1 - \cos x)(1 + \cos x)(1 + \sqrt{\cos x} \cdot \sqrt{\cos 2x})} = \frac{5}{4} \quad \dots \quad \cos x$$

$$16) \lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} = \lim_{x \rightarrow 1} (1-x) \cdot \cot \left(\frac{\pi}{2} - \frac{\pi x}{2} \right)$$

$$= \lim_{x \rightarrow 1} \frac{1-x}{\tan(\frac{\pi}{2} - \frac{\pi x}{2})} = e^{u \cdot (u-1)}$$

$$17) \lim_{x \rightarrow \infty} \left(\frac{3x+1}{3x+2} \right)^{4x} = \lim_{x \rightarrow \infty} e^{4x \left(\frac{3x+1}{3x+2} - 1 \right)}$$

$$= \lim_{x \rightarrow \infty} e^{\ln\left(\frac{-1}{3x+2}\right)} = e^{\frac{-4}{3}}$$

$$18) \lim_{x \rightarrow \infty} \left(\frac{3x^2 + 1}{3x^2 + 5} \right)^{2x^2 + x} = \lim_{x \rightarrow \infty} e^{(2x^2 + x) \left(\frac{-4}{3x^2 + 5} \right)} = e^{\frac{-8}{3}}$$

$$21) \lim_{x \rightarrow 1} (1 + \sin \pi x)^{\cot \pi x}$$

$$\lim_{x \rightarrow 1} (1 + \sin \pi x) = 1$$

$$\lim_{x \rightarrow 1} \cot \pi x = \lim_{x \rightarrow 1} \frac{\cos \pi x}{\sin \pi x} = \frac{\cos \pi}{\sin \pi} = \frac{-1}{0} = -\infty$$

$$\Rightarrow \lim_{x \rightarrow 1} e^{\cot \pi x \cdot \sin \pi x} = \lim_{x \rightarrow 1} e^{\cos \pi x} = e^{-1}$$

$$22) \lim_{x \rightarrow 0} (1 - 2x^2)^{\cot^2 x}$$

$$\lim_{x \rightarrow 0} (1 - 2x^2) = 1$$

$$\lim_{x \rightarrow 0} \cot^2 x = \lim_{x \rightarrow 0} \frac{\cos^2 x}{\sin^2 x} = +\infty$$

$$\Rightarrow \lim_{x \rightarrow 0} e^{-2x^2 \cdot \cot^2 x} = \lim_{x \rightarrow 0} e^{-2x^2 \cdot \frac{1}{x^2}} = e^{-2}$$